

Mid Semester Examination 2025

Name of the Program: B.Tech Petroleum Engineering

Semester: VI

Name of the Course: Alternate Energy Resources

Course Code: TPE-611

Time: 1-1/2Hour

Maximum Marks: 50

Note:

(iii) Answer **all the questions** by choosing *any one of the sub questions*.

(iv) Each question carries 10 marks

Q1	(10 marks)	CO 1
(a)	Discuss in details about the non-conventional energy sources and their advantages and estimated potential?	
OR		
(b)	Define fuels? What are the classification fuels? Discuss in detail about the solid fuels	
Q2	(10 marks)	CO 2
(a)	What are the renewable energy resources? Discuss advantages and prospects of renewable energy resources	
OR		
(b)	Write a short note on a sunshine recorder? Explain their working and construction with labelled diagram?	
Q3	(10 marks)	CO 2
(a)	Define the following: a. Latitude angle b. Hour angle c. Diffuse radiation d. Concentration ratio e. Clarity index	
OR		
(b)	Discuss the solar radiation geometry specifying the different angle used in energy calculations with the help of diagrams?	
Q4	(10 marks)	CO 1 CO 2
(a)	Define the characteristics features of a collector system? Discuss the factors adversely affecting the collectors system's efficiency	
OR		
(b)	Discus in details about the types of collectors? Explain the performance analysis of flat plate collectors (FPC)	
Q5	(10 marks)	CO 2
(a)	Discuss the classification solar energy storage systems? Hence explain the basic equation for the energy storage unit?	
OR		
(b)	Illustrate the following storage a. Battery storage b. Capacitor storage c. Inductor storage	

